

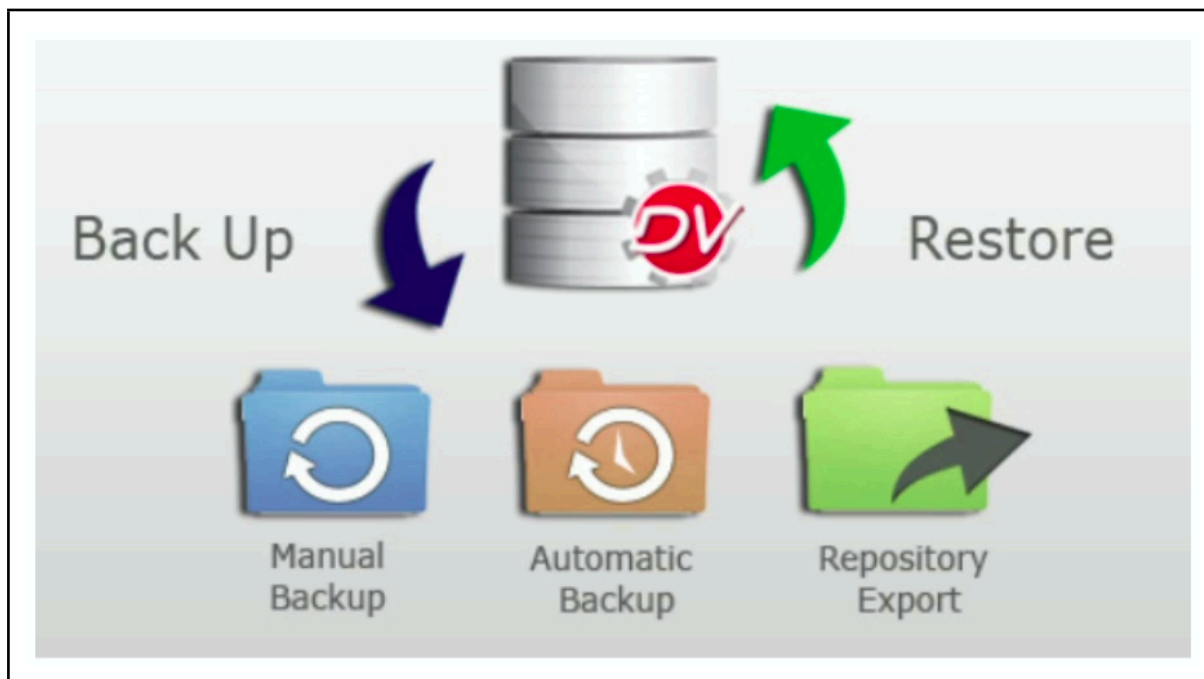
Section 7: System Performance

Backup & Recovery Strategies

In this lecture we will explore how effective backup and recovery strategies protect data, minimize downtime, and ensure that systems can quickly recover from failures and disasters. So let's get started.

What is Backup & Recovery?

One common misconception is that replication alone is enough to protect your data, which is not. If bad data gets replicated or ransomware encrypts your files, every replica can end up with the same problem and that is where backup and recovery becomes very essential. A backup creates a separate point-in-time copy of your data, while recovery is the process of restoring that data when something goes wrong. Together, they form your last line of defense against accidental deletion, corruption, hardware failures, and even large-scale disasters.



In other words, replication keeps your services available, but backup and recovery ensure that your data can actually be brought back when everything else fails. A resilient architecture needs both working together.

Why is Backup Important?

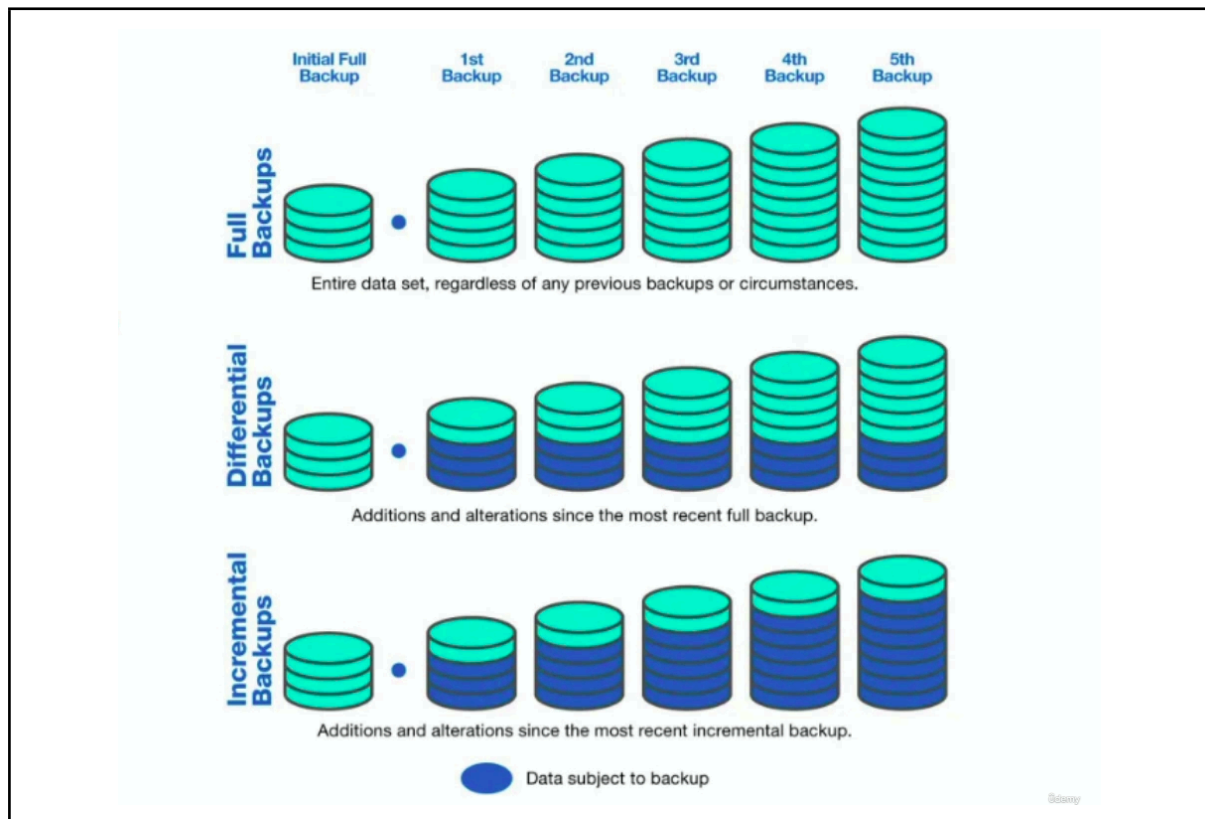
Backups are important because failures are inevitable. It is only a matter of when and not if. Disks fail, software bugs corrupt data, and entire servers can suddenly become unavailable. Without a backup, that data may be gone for good. Not every incident is caused by technology. Human error is one of the most important reasons for data loss. An accidental delete, an incorrect update, or a misconfigured script can wipe out critical information in seconds.



Then there are external threats. Ransomware can encrypt data, and natural disasters like fire, flood, or power outages can take entire data centers offline. In both cases, backups are often the only reliable way to recover.

Finally, backups are not just about recovery, they are also about compliance. Many industries require organizations to retain data for months or even years. A well designed backup strategy helps to meet those legal obligations, while ensuring that the business can recover from unexpected failures with confidence.

Types of Backup

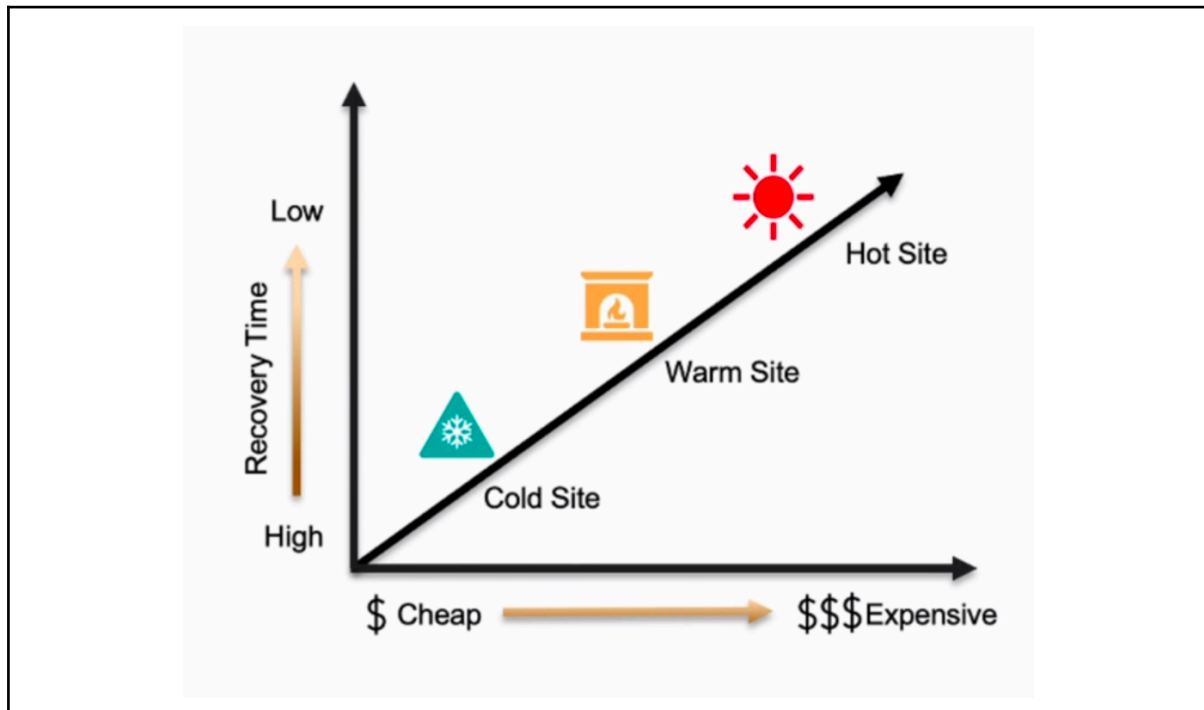


When designing a backup strategy, you are always balancing 3 things: storage cost, backup time and recovery speed. That is why different types of backup exist, each optimized for a different priority.

- **Full Back:** This is the simplest approach because it copies everything every time. Recovery is straightforward since you only need a single backup set, but the tradeoff is higher storage consumption and longer backup windows.
- **Incremental Backup:** This type of backup only stores the changes since the most recent backup. Whether that was a full or another incremental backup doesn't matter. This makes backup much faster and more storage efficient. However, recovery is more complex because you need the last full backup along with every incremental backup that is created after it.
- **Differential backup:** It stores all the changes made since the last full backup. As days pass, differential backups grow large, but recovery is simpler. You only need the last full backup and the latest differential backup.

In the production system, it is very common to combine these approaches. For example, a weekly full backup with daily incremental backups provide a practical balance between storage efficiency, backup performance, and recovery time.

Recovery Types



Recovery is not just about restoring the data. It is also about quickly your business can resume the operations. The right recovery strategy depends on how much downtime your organisation can tolerate and how it is willing to invest.

- **Cold Recovery:** With cold recovery, backups are stored safely, but the infrastructure is not ready. After a failure, servers, database, and applications must be rebuilt before data can be restored. It is the most cost-ineffective option, but it is also the one that results in the longest downtime.
- **Warm recovery:** This recovery provides a middle ground. Some infrastructure is already provisioned and can be brought online very quickly, while the remaining components are restored from backups. This reduces recovery time without the cost of maintaining a full active standby environment.
- **Hot Recovery:** This recovery has a fully synchronized secondary environment that is always running and ready to take over immediately. This delivers downtime and is a preferred choice for mission-critical systems, but it also has the highest infrastructure and operational cost.

Ultimately, choosing between cold, warm, and hot recovery is a business decision. The more downtime a system can tolerate, the lower the recovery cost. For systems where every minute of outage translates into significant financial or operational impact, investing in hot recovery is often justified.

Understanding RTO & RPO



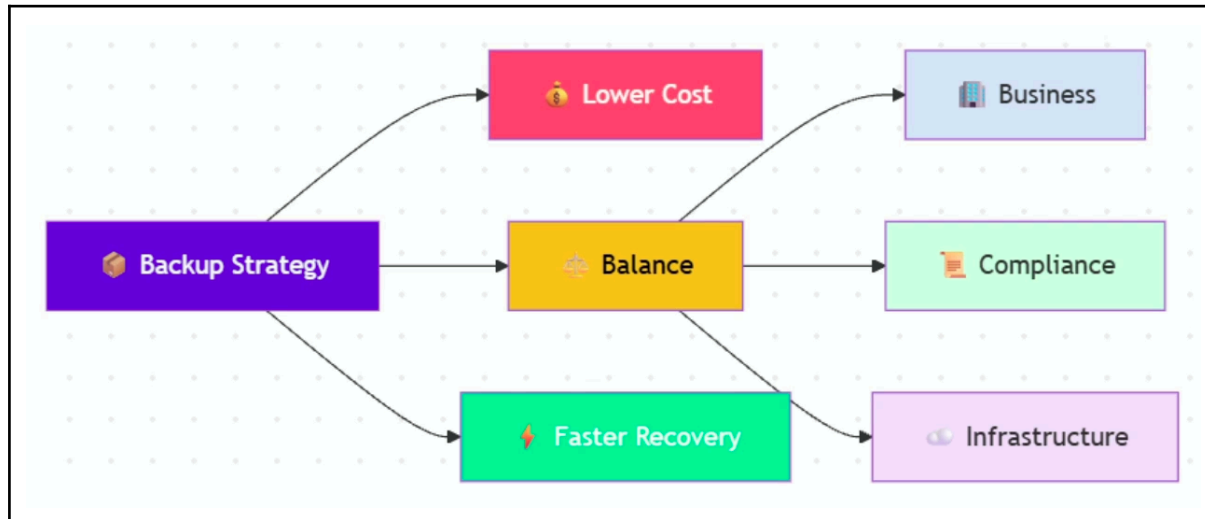
RTO & RPO are two of the most important metrics used in disaster recovery as they define what successful recovery actually means from a business perspective. They answer two different but equally important questions.

- **RPO (Recovery Time Object):** This is about time. It defines how quickly your systems must be restored after an outage. If the business can only tolerate one hour of downtime, your architecture, automation, and recovery process must be capable of meeting that target.
- **RPO (Recovery Point Target):** This is about data. It defines the maximum amount of data that you are willing to lose. For example, an RPO of 15 minutes means backup or replication must ensure that, at worst, only the last 15 minutes of data could be lost.

These 2 objectives directly influence your architecture. Achieving shorter RTOs and RPOs typically require more frequent backups, faster recovery mechanisms, additional infrastructure, and greater automation. That is why architects don't use RTOs and RPOs based on technology. They start with business requirements and then design a solution that meets those objectives at an acceptable cost.

Trade-offs in Backup Strategy

There is no single backup strategy that works for every system because every decision involves tradeoffs. As an architect, the goal is not to eliminate the tradeoff, it is to make the right ones for the business.



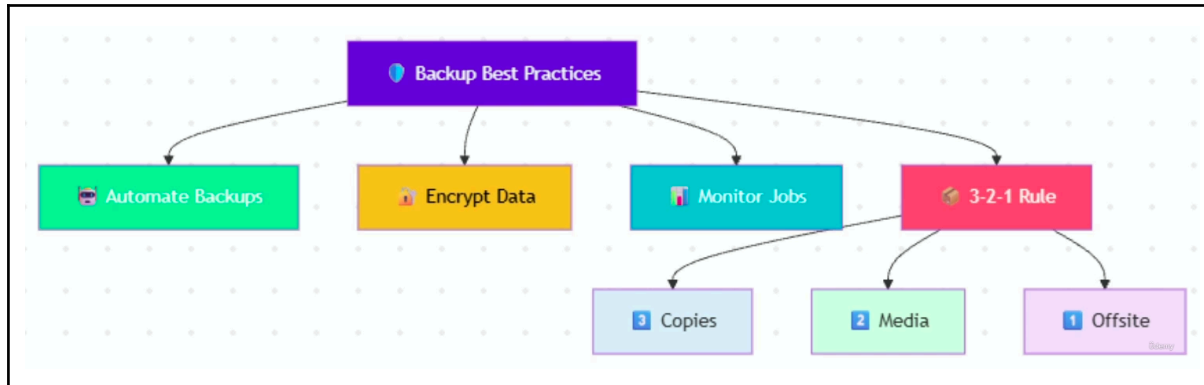
The first tradeoff is cost vs. recovery vs. speed. Faster recovery usually means more frequent backups, additional infrastructure, or standby environment, all of which increase operational cost. A lower cost solution may be perfectly acceptable if the business can tolerate longer recovery times.

Another consideration is the complexity of managing backups over time. Deciding how often to backup data, how long to retain it, and how to verify that backups are actually restorable becomes increasingly challenging when your systems grow.

Ultimately, your backup strategy should align with business priorities. Critical customer-facing systems often justify higher investment. Compliance requirements may dictate retention policies, and the maturity of your infrastructure determines how much automation and operational complexity your team can realistically manage. The best strategy is the one that delivers the required level of protection without adding unnecessary cost or complexity.

Best Practices

A backup strategy is only valuable if it works reliably when you need it. That is why experienced teams focus not just on creating backups, but on making the entire backup process dependable, secure, and continuously validated. Start by automating both backup and restore testing.



Automating backups eliminates manual errors, while regularly testing restore confirms that your backups are actually usable. A backup that cannot be restored is effectively no backup at all. Security is equally important. Backups should be encrypted, both at rest and in transit, and every backup job should be monitored.

Successful backups, failures, and storage capacity should all generate alerts so that issues are detected before they become real incidents. A widely accepted guideline is this 3-2-1 rule. Maintain 3 copies of your data, store them on at least 2 different types of storage, and keep one copy off-site. This simple strategy provides protection against hardware failures, accidental deletions, ransomware, and even site-wide disasters.

Ultimately, the goal is not just to create the backups. It is to ensure that your organization can recover quickly, securely, and with confidence whenever there is a failure.

Summary & Key Takeaways

- Backups are vital for disaster recovery and resilience
- Choose the right backup type & recovery model
- Understand RTO/RPO goals
- Cloud solutions simplify but require governance
- Regular testing is key!
- What's next:
 - Disaster Recovery in Practice

Interview Questions

- What is the difference between full, incremental, and differential backups? When would you use each?
- How do you define and balance RTO and RPO in a large-scale distributed system?
- Explain cold, warm, and hot recovery strategies with examples
- How would you implement a backup strategy for a microservices-based application hosted in the cloud?
- What trade-offs do you consider when designing a backup and recovery system for a high-availability service?
- How does cloud storage simplify or complicate backup and recovery strategies?
- What are some best practices for backup automation and testing in production systems?
- How would you handle backup for a database with terabytes of data and minimal allowed downtime?